



Infor Mongoose

Guide to Technology

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Publication Information

Release: Infor Mongoose 9.03.12

Publication Date: 31 August 2017

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Use of this document

This document is intended for readers who understand basic network and infrastructure design models. The document leaves out some detail in order to provide a generalized technical document. Please read this document in its entirety.

This document is intended to provide information on the infrastructure deployment of Infor solutions. This document does not take into account all scenarios, infrastructure challenges, and solution options. We recommend that, if your intent is to purchase hardware or infrastructure to support or modify an Infor solution, you should contact your sales representative. The Infor sales representative can then get the appropriate resource involved to review the technical and infrastructure recommendation and possibly provide a specific configuration recommendation. This document is confidential and property of Infor Global Solutions, Inc. Unauthorized use of this document is strictly prohibited.

Infor Mongoose

Overview

Infor Mongoose is a software development framework. It provides for rapid development of applications that use relational database technology. Mongoose is a core Infor technology, providing built-in support for other Infor core technologies, such as Infor Ming.le and Infor ION. Infor SyteLine has been built on Infor Mongoose since its version 7 launched in 2002.

The central defining goal of the Infor Mongoose framework is to create feature-rich applications while minimizing the amount of time and program coding required. Therefore, Mongoose maximizes the assets companies invest in IT infrastructure and provide end users the ability to adapt Mongoose-built applications quickly and cost effectively to react and adapt to emerging business requirements without sacrificing the ability to upgrade the application to newer versions.

By minimizing the need for program coding in adapting the Mongoose-built applications, changes are automatically upgradeable, development is accelerated, quality is increased through extreme code reuse, and your investment is protected and insulated from technology changes. Since creating *some* new code is inevitable, Mongoose streamlines the process by enabling your team to use familiar and standard Microsoft .NET technologies and tools.

Together with other factors, such as simplicity of installation and configuration, support for a wide range of integrations, a rich user interface for end user productivity, an automatic browser user interface, multi-device features, and cloud or on-premise deployment options, this makes Infor Mongoose an unparalleled value.

To meet these objectives, Infor Mongoose uses a model defined by metadata (data defining the application behavior, stored in rows and columns in databases) for the user interface (forms), the mid-tier business objects (Intelligent Data Objects, or IDOs) and business processes (Application Event System, or AES). Infor Mongoose provides metadata-driven constructs and patterns to build your application. Therefore, developers and customizers do not need to worry about building them. In summary, Infor Mongoose provides:

- **A single developer experience.** Develop and test in the same integrated environment. Core forms and wizards let you define Intelligent Data Objects (IDOs) that represent your business logic and application events (AES) without programming. A simple click turns Infor Mongoose into a powerful form development tool.
- **Software development life cycle.** Version control, development, test, preproduction, production environments and synchronization tools help you manage and organize the entire software development life cycle and deployments.
- **Form templates.** Mongoose provides for a consistent user experience across the application.
- **Prebuilt common objects.** Mongoose has, built into it, UI components, validations, standard events, messages, menus, variable support, scripting APIs, user-defined fields, and more.
- **Internationalization.** Mongoose provides easy translation of string literals based on the end user's default language, and the ability for end users to switch languages "on the fly" without restarting the application.
- **Security.** Security is provided at user, role, and site-based levels for all forms and IDOs.

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- **Licensing.** System and end users can be assigned licensing on a variety of modules.
 - **Documents and notes management.** A virtually unlimited number of notes and documents can be attached to any data.
 - **Data and process replication.** Mongoose supports customer policy for both transactional and asynchronous XML options for sites in both your local intranet and across the Internet.
 - **Easy and powerful integration with other software.** Exchange information with other software using Infor ION, Web services, XML, or another application's Web UI embedded in a form.
 - **Background task execution.** You can schedule various tasks, such as report outputs and notifications, running stored procedures, and other tasks to run automatically in the background.
 - **Reporting interfaces.** Mongoose includes a number of prebuilt report options, as well as a set of templates for creating your own reports.
 - **Customer extensibility.** The same powerful integrated development environment is available to end users of Mongoose-built applications to extend and adapt them. More importantly, all the changes are preserved during upgrades easily, using synchronization tools.
 - **Source control support.** Mongoose integrates with Microsoft Visual Source Safe, Microsoft Team Foundation Server, and Apache Subversion for the purpose of maintaining metadata change histories and source version control.

Supporting technology

Operating systems – servers

Microsoft Windows Server 2012 / 2012 R2 / 2016
Standard or Enterprise Edition, 64-bit

It is important to understand the differences in editions, as features in Standard and Enterprise editions vary and might be desirable. For instance, there are limitations in the Standard versions that are not present in Enterprise releases. Currently, there is no official support for Itanium 64-bit operating systems.

Database technology

Microsoft SQL Server 2012 | 2014 | 2016

- Standard or Enterprise Edition, 64-bit
- SQL Server Reporting Services installed

It is important to understand the differences in editions, as features in Standard and Enterprise editions vary and might be desirable. For instance, there are limitations in the Standard versions that are not present in Enterprise releases.

Infor is a reseller of Microsoft SQL Server. Contact your sales representative if you need to purchase SQL server for your deployment.

For additional information, see Microsoft's Web site at www.microsoft.com.

Mongoose clients

It is important to understand the type of clients that are available with Mongoose 9.

Client types for Mongoose include:

- **Smart Client in Windows**
 - Client that includes the ability to run in Design Mode
 - Requires a supported Windows Operating System with .NET framework
 - Deployment method: Installed using Click-Once and updated automatically
 - Commonly used across LAN, WAN, and Internet connections
- **Web Client**
 - HTML5 client that can run in common browsers on popular devices
 - Commonly used across LAN, WAN, and Internet connections
 - Used to access Mongoose in the cloud

Mongoose Smart Clients supported platforms

- Microsoft Windows 8.1 Professional
- Microsoft Windows 10 Professional
- Microsoft Server 2012
- Microsoft Server 2012 R2
- Microsoft Server 2016
- Citrix Server add-on
- Browsers for Mongoose – Internet Explorer, Firefox, Chrome, and Safari

Mongoose Web Client supported platforms

- Device agnostic – Internet Explorer, Firefox, Chrome, and Safari
- Mobile via browser– Android, iPhone, Blackberry

Deployment

DMZ – The “DeMilitarized Zone”

There are several deployment options available for Infor Mongoose-based applications and supporting components. These include certain scenarios where the framework and application components might be exposed to a public network (the Internet, in most cases) for access to users or other systems. This perimeter network is usually a physical or logical subnetwork that contains and exposes external services to the public network. Access between this DMZ subnetwork and the Internet, for example, are usually limited to only certain ports and methods of access. Additionally, there are rules for the server or services that are residing in the DMZ to access systems that are on the organization’s private network. This limits the amount of system exposure to public attacks.

Examples and features

Examples of Infor Mongoose use cases or applications that might be accessed through the DMZ include but are not limited to:

- Windows server with Internet Information Services (IIS)
 - “Smart client” (WinStudio) users
 - Web client users
 - Mobile users
 - Web services requests

One feature included in Mongoose is the DMZ server installation option. A server that is installed in the DMZ could be used for this functionality. The installation of the Mongoose DMZ server includes the IDO Request Service and Web client without the full utility server components. You would choose this option if you do not want to expose your utility server with all the utility server components into your DMZ.

In all scenarios, verify that your Microsoft Server and Microsoft SQL Server licenses are correct for the type of user and data you are planning to expose.

The Mongoose DMZ Server

Below is an example of a DMZ server specification. This specification generally should support 100 or more concurrent Web-rendered or mobile users. The count is for this server and the concurrent count still needs to be considered for the utility server with the IDO Runtime service that is on the private network.

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 4-8 (must be expandable)
Memory size	➤ 12 GB
Disk I/O configuration	➤ Two 72 GB SCSI / SAS drives, RAID 1
Other	➤ Internet Information Server (IIS)

	➤ Mongoose DMZ framework
Network interface	➤ 1 GB ethernet adapter

Virtualization

For Mongoose release 9.00 or later, Infor supports customers running a production Mongoose solution suite in a Microsoft Hyper-V or VMware virtual environment. It is expected that the customer is running the guest virtual machines and components using the supported technologies defined for the specific release of the Mongoose solution. The Mongoose solution functions in a virtual environment (also known as “hypervisor”) as if running on physical native hardware. If Infor Mongoose Support suspects that the virtualization layer is the root cause of an error or issue, you are required to contact the virtualization solution provider to resolve the virtualization issue.

Virtualization sizing

With the support of virtual environments for Infor Mongoose-based applications, it is your responsibility as a customer to work directly with the hypervisor provider of choice to ensure that the sizing of the host server hardware, versions, configuration, setup, and device mappings are correct. Because of the number of variables that can be introduced into such an environment, Infor does not provide hardware sizing for virtualization environments. You should work closely with your hypervisor provider to ensure that the correct hardware sizing and configuration are selected to match your requirements.

Other virtualization solutions

Virtualization is a growing segment of the technology market. Infor has chosen Microsoft Hyper-V and VMware as the only virtualization solutions that are officially supported for Infor Mongoose. There are many other virtualization solutions in the market from providers like Citrix and RedHat. Using other virtualization solutions still allows you to receive support for Infor Mongoose, but you should read and understand this *Non-Standard Software Deployment* statement concerning Infor Mongoose:

Support Policy for Infor Mongoose Non-Standard Software Deployments

Infor Mongoose is designed, developed, tested, and documented to run in a multi-tier/multi-server deployment using various software components at specific versions and service pack levels. These components include such things as the operating system (OS), databases, and other components as listed in the technical documentation provided with each Infor Mongoose release. This document addresses support provided by Infor for its software running in configurations that do not conform to the standard documented software deployment.

Infor does not test or maintain any support environments of Infor Mongoose software running in non-standard software deployments. Therefore, Infor will require any support issue reported to be reproduced in a standard software deployment system, before an APAR (software defect) will be logged. Where the issue is confirmed to be unrelated to the non-standard software deployment, Infor will support its software in a manner that is consistent with support provided when that software is running on a standard software deployment.

Infor will provide code corrections, according to the Infor Mongoose On-Demand Patch Policy, which are developed and tested in systems that are running on a standard software deployment. These code corrections will be made available by means of the normal code correction systems and procedures used by all customers running Infor Mongoose.

Customers running non-standard software deployments of Infor Mongoose can also access and apply these code corrections using standard procedures. Infor makes no warranty, implied or otherwise, regarding the use of these fixes in non-standard software deployments.

Additional information about non-standard software deployments

Customers most often request non-standard software deployments when they order new hardware with software components that are more advanced than the software on the approved deployment list, and they cannot “back level” the software to the standard versions. Another common occurrence is that a non-Infor software vendor has suggested applying one or more fixes to one of the non-Infor software components, and this requires installing a service pack level that is higher than the one on our standard software deployment list. In most cases, service packs or newer versions of these components will support all of the functions and commands found in the currently supported version. For this reason, the risk of installing the newer code is small. However, we have seen situations where new versions or service pack levels have caused issues that have impacted Infor Mongoose and required the new software to be removed.

Specific information regarding normal software deployment requirements is available from our Infor Xtreme Support Web site (<http://www.inforxtreme.com>) > **Documentation** > **Mongoose Framework** product line. Infor support can update or change the Standard Software Deployment list as new versions of software are available and can be tested internally.

Since the support teams only have access to standard documentation and systems running in standard software deployment configurations, we are unable to assist with specific non-standard software deployment installation, setup, or performance questions and recommendations. Any and all information related to setting up a non-standard software deployment is solely your responsibility as a customer. This information can be obtained from various sources which are not part of Infor support, and can result in successful deployment of the software for a specific customer environment.

You, the customer, assume all risks associated with conflicts that arise between various system component software including, but not limited to, performance and system settings.

Infor software might not work as intended in non-standard software deployments, because of unknown conflicts between Infor Mongoose and the new software version or component. If these issues cannot be resolved by means of normal code fixes, you might have to implement a standard software deployment system to correct the issue.

This information is subject to change, without notice, at any time.

Sizing for Infor Mongoose-based applications

Important Notice:

If you plan to deploy in the Web, then when sizing the utility server, treat one session as three concurrent sessions. For example, a utility server configured for 300 concurrent users of a Mongoose-built application using WinStudio (the smart client) supports only 100 Web-rendered clients instead. The reason for this additional overhead on the server is that the rendering of forms and form scripts, which is otherwise done by WinStudio, must be handled by the server in a Web-rendered presentation. Both WinStudio and the Web-rendered presentation methods work across LAN, WAN, and Internet connections. This sizing model for Web rendering would only apply to the utility server load if the DMZ server is not used.

Each user running over a WAN connection consumes approximately 50Kbps to 100Kbps of bandwidth. The recommended Wide Area Network (WAN) configuration is 50Kbps per active user (50Kbps x 10 users = 500Kbps).

All sizing information for a utility server in this guide is intended for WinStudio connectivity to the utility server and uses the leading Mongoose application, Infor ERP Business (SyteLine), as a parameter to guide you.

For sizing your development environment, consider 1-20 and 21-50 concurrent users for combined servers.

Size and configuration recommendations

Note that the recommendations in this section are *only* recommendations. You might need to conduct level and performance evaluations to determine the configuration for your specific needs.

1-20 Concurrent Users

Database/Utility Servers (combined servers)

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 4 cores
Memory size	➤ 12 GB (must be expandable)
Disk I/O configuration	➤ (6) 72/144/300 GB SCSI / SAS Drives
Other	➤ Terminal Server Remote Administration
	➤ SQL Server 2012 2014 2016
	➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

21-50 Concurrent Users

Database/Utility Servers (combined servers)

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 16 GB (must be expandable)
Disk I/O configuration	➤ (6) 72/144/300 GB SCSI / SAS Drives
Other	➤ Terminal Server Remote Administration ➤ SQL Server 2012 2014 2016 ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

51-100 Concurrent Users

Database Server

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 20 GB (must be expandable)
Disk I/O configuration	➤ (8) 72/144/300 GB SCSI / SAS Drives
Other	➤ Terminal Server Remote Administration ➤ SQL Server 2012 2014 2016 ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

Utility Server

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores

Memory size	➤ 12 GB
Disk I/O configuration	➤ (2) 72 GB SCSI / SAS Drives RAID 1
Other	➤ Internet Information Server (IIS) ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

101-200 Concurrent Users

Database Server

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 24 GB (must be expandable)
Disk I/O configuration	➤ (10) 72/144/300 GB SCSI / SAS Drives
Other	➤ Terminal Server Remote Administration ➤ SQL Server 2012 2014 2016 ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

Utility Server

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 16 GB
Disk I/O configuration	➤ (2) 72 GB SCSI / SAS Drives RAID 1
Other	➤ Internet Information Server (IIS) ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

201-300 Concurrent Users

Database Server

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 48 GB (must be expandable)
Disk I/O configuration	➤ (12) 72/144/300 GB SCSI / SAS Drives
Other	➤ Terminal Server Remote Administration ➤ SQL Server 2012 2014 2016 ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

Utility Servers (1 or 2)

Operating system	➤ Windows Server 2012 / 2012 R2 / 2016
Processor size	➤ Xeon 2.0 GHz (or better)
Number of processors	➤ 8 cores
Memory size	➤ 16 GB
Disk I/O configuration	➤ (2) 72 GB SCSI / SAS Drives RAID 1
Other	➤ Internet Information Server (IIS) ➤ Mongoose-built application code
Network interface	➤ 1 GB ethernet adapter

300+ Concurrent Users

It is possible that a customer end-user base could include more than 300 concurrent users. We recommend that you complete some level of performance evaluation for any application you create with Infor Mongoose and adjust your system configuration to meet your needs.

User access

A Smart Client provides the fully functional and user-friendly graphical interface of desktop applications with the communication protocol http/http(s) to allow internet or WAN deployment. The client leverages “click-once” deployment for ease of installation and is auto-updating from the Mongoose servers.

Operating systems	➤ Windows 8.1 Professional ➤ Windows 10 Professional
Processor size	➤ Multi-core, 1.6 GHz (or better)
Number of processors	➤ 2-4 cores
Memory size	➤ 2/3 GB (depending on the OS)
Other	➤ .NET 4.6.1 framework ➤ Network connection

Web client (Internet browser-based)

Forms that are supplied with the Mongoose-based application, whether personalized or created from scratch, can be rendered in a Web browser after the forms are saved and permissions are assigned. These browsers are supported for form Web rendering:

- Internet Explorer
- Firefox
- Safari
- Chrome

Mobile device Web browsers

Mobile phones and devices that have a Web browser can use Mongoose Web-rendered mobile forms that are supplied with the Mongoose-built application, whether personalized or custom-created.